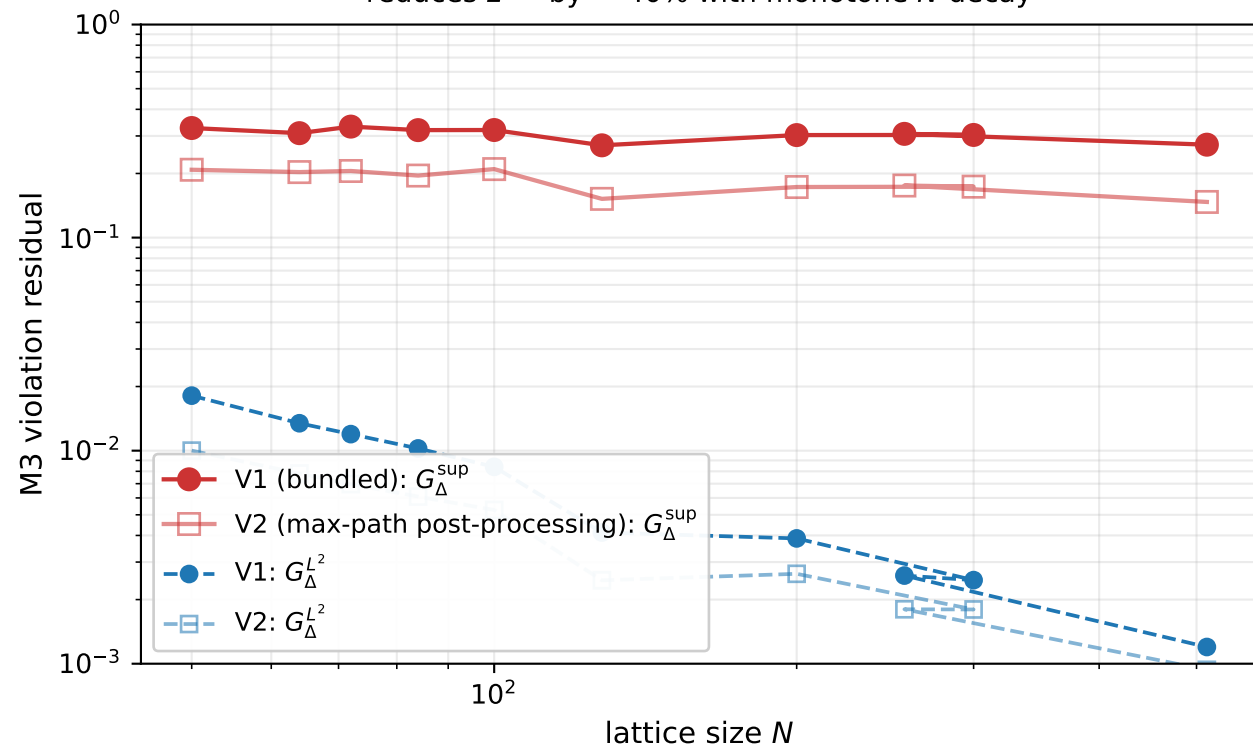


M3 sub-multiplicative-triangle violations on the within- P_5 multi- N ladder

(a) Multi- N M3 violation: L^2 converges, L^{sup} stays bounded under V1; V2 envelope refinement reduces L^{sup} by $\sim 40\%$ with monotone N -decay



(b) Heavy-tail correlation: M3 sup-norm violations are matter-cluster-localised at 2--3 $\times$ above random null

